

DE 424: Wumpus Trajectory Tracking Model

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May 13, 2026

Declare that this work presented is your own work. Mention any sources that contributed to work or report
I declare that this report is my own work. This report was written using LaTeX and thus Gemini was used to assist in certain formatting. This includes help to format code listings, mathematical equations and image display. The mathematical derivations and interpretation of results are my own work.

ECSA graduate attributes

Discuss how the GAs were assessed in the module + which parts of the report this is documented. State HOW and WHERE they were satisfied This section may (and should) be brief, but should discuss all aspects of the GAs being assessed. For example, GA1 partially assesses your ability to research literature as part of problem solving, which means that your report must mention and cite (at least) a few sources - can include prescribed resources. Please consult the module framework for a more detailed description of these two GAs as well as how they are assessed in this module.

GA 1: Problem solving

GA 2: Application of scientific and engineering knowledge

1 Problem Description

This project addresses an object tracking task within a discrete grid world inhabited by a latent entity known as a wumpus. The objective is to determine the trajectory of the wumpus across multiple time steps based on inaccurate measurements. At each time step the wumpus attempts to move to an adjacent cell (up, down, left, or right) with equal probability. If the target cell is within the grid boundaries, the wumpus transitions to that location; otherwise it remains in its current position. The exact location of the wumpus is never directly observed. Instead, every cell in the grid generates a binary detection value at each time step, parameterised by:

- True Positive (p_w): The probability that the cell containing the wumpus correctly generates a detection.
- False Positive (p_c): The probability that a cell not containing the wumpus generates an incorrect detection (clutter measurement).

Given the stochastic grid detection data across multiple time steps, the task is to design, implement and evaluate a Probabilistic Graphical Model (PGM) solution. This involves constructing a model that accurately captures the wumpus's transition and observation logic to infer its latent trajectory across various datasets.

2 Solution Design

The solution was developed iteratively to accommodate the increasing complexity and varying characteristics of each dataset. The PGM design process followed a structured methodology: defining the random variables, defining the model, and constructing the model factors. This process is detailed below for each attempted dataset. (note: this section must show model structure and show all factors. State and motivate all design choices)

2.0.1 Dataset 1

The initial model was designed for a 5×5 grid over a series of $T = 10$ time steps. In this case, the parameters $p_w = 0.95$ and $p_c = 0.05$ are known constants.

2.1 Random Variable Definition

A critical design choice was made to represent the wumpus’s location using a single latent RV, W^t , instead of separate X^t and Y^t coordinates. The grid is mapped to a 1-dimensional vector where each cell index $i \in \{0, \dots, 24\}$ corresponds to a unique (x, y) coordinate. This approach offers several advantages. Primarily, it enhances inference efficiency by reducing the total number of RVs within the cluster graph, which leads to more efficient message passing and overall faster convergence. Furthermore, it reduces complexity by simplifying factor construction and avoiding the complexities of coordinating the dependencies between spatial coordinates. The wumpus’s movement constraints are more easily enforced within a transition matrix; for example, the probability of illegal moves, such as diagonal steps, can be explicitly set to zero within one factor. The RVs are defined as follows:

- $W^t \in \{0, 1, \dots, 24\}$: The latent position of the wumpus at time t .
- $D_i^t \in \{0, 1\}$: The binary detection status of cell i at time t , where 1 indicates a detection.

2.1.1 Model Definition

The wumpus tracking task is formulated as a Hidden Markov Model (HMM), a framework used to infer a sequence of hidden states from observable outputs [1]. The latent trajectory of the wumpus ($W^0, W^1, \dots, W^T$) is modeled as a first-order Markov process, $P(W^t | W^{t-1})$, where every future state depends on the current state, not on the sequence of states preceding it. The observable detection variables are emissions that are generated from each state $p(D_i^t | W^t)$. A HMM is a subclass of Bayesian Networks, used for sequential data. The model is illustrated below in Figure 1.

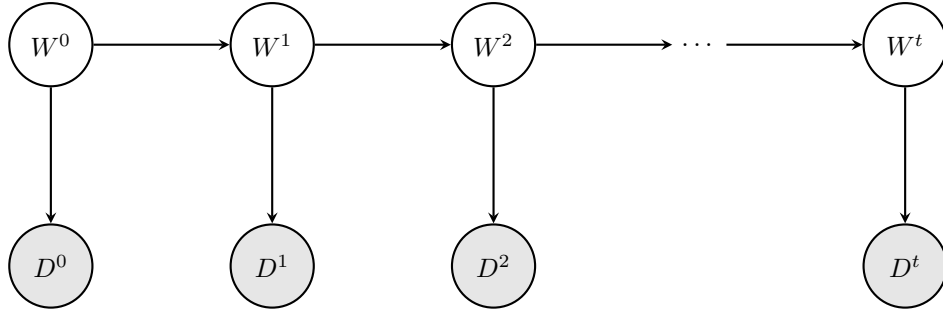


Figure 1: Bayesian Network structure of the HMM tracking solution.

2.1.2 Factor Definition

The model is governed by two primary factors. The first factor models the latent movement

Two factors are defined in this model. Firstly, one to model the motion of the wumpus over time, $p(W^t | W^{t-1})$. This factor captures the wumpus random movements to adjacent cells. The wumpus can only move up, down, left and right, each having a probability of 0.25. If it attempts to leave the boundary of the grid, it will remain in place. Due to this, if the wumpus is on a cell bordering the boundary, it will have a 0.25 prob of remaining in place, and if wumpus is in a corner (can’t move to two out of four directions), it has a prob of remaining in place = 0.5. This stochastic time dependent trajectory can be modelled by a transition matrix. In the case of a 5x5 grid, the resulting transition matrix is 25x25.

Secondly, the other to model the probability of a cell detection conditioned on the wumpus location $p(D_i^t | W^t)$.

Table 1: Illustrative sample of the Transition Probability Factor $P(W^t | W^{t-1})$ for a 5×5 grid.

W^{t-1}	W^t	$P(W^t W^{t-1})$	Description
0	0	0.50	Stay (Blocked Up and Left)
0	1	0.25	Valid Move (Right)
0	5	0.25	Valid Move (Down)
0	6	0.00	Illegal Move (Diagonal)
12	12	0.00	Stay (No Boundaries Hit)
12	13	0.25	Valid Move (Right)
...	...	...	...

Table 2: Illustrative sample of the Observation Factor for a single cell i at time t .

W^t	D_i^t	$P(D_i^t W^t)$	Description
i	1	p_w (0.95)	True Positive (Correct Detection)
i	0	$1 - p_w$ (0.05)	False Negative (Missed Detection)
$j \neq i$	1	p_c (0.05)	False Positive (Clutter)
$j \neq i$	0	$1 - p_c$ (0.95)	True Negative (Correct Non-Detection)

Furthermore, because the resulting structure is a junction tree, the algorithm is guaranteed to reach exact inference in only two passes (forward and backward), ensuring maximum computational efficiency and faster overall runtime.

3 Code Implementation

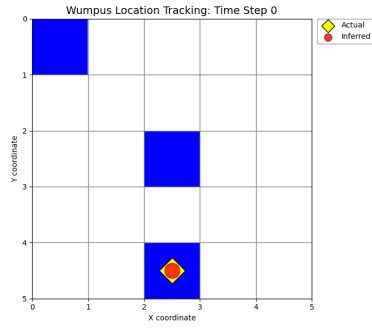
Discussion + code snippets

4 Analysis of Results

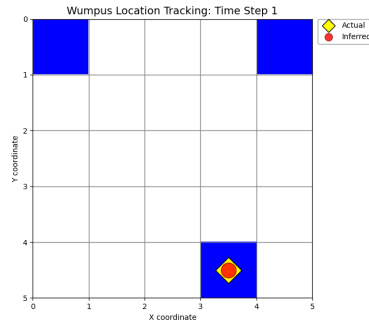
Description of how you processed the output of solution, the presentation of results and analysis thereof. Report both qualitative and quantitative results.

References

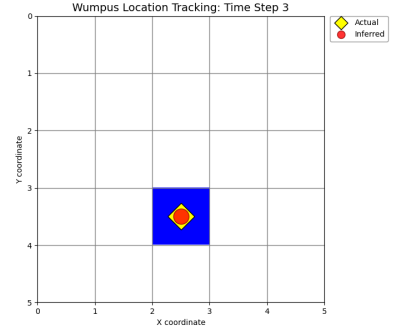
- [1] Daniel Jurafsky and James H. Martin. *Hidden Markov Models*. 2023. Appendix A.



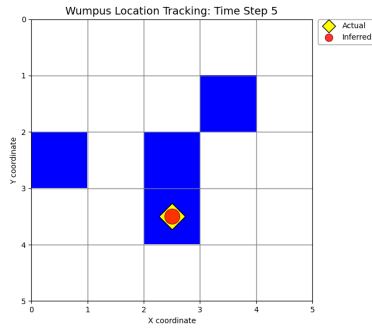
(a) Time $t = 0$



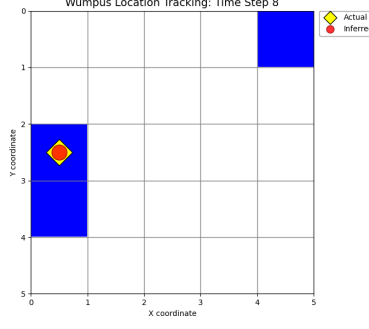
(b) Time $t = 1$



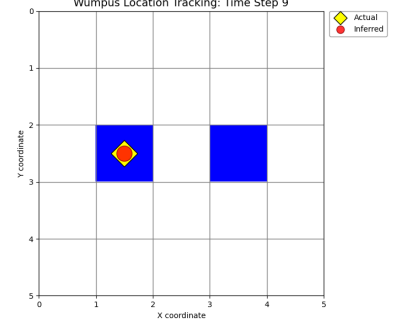
(c) Time $t = 3$



(d) Time $t = 5$



(e) Time $t = 8$



(f) Time $t = 9$

Figure 2: Dataset 1 tracking performance across select time steps. The grid displays binary detection data, where the **red dot** represents the ground-truth wumpus location and the **yellow star** represents the location inferred by the MAP algorithm.